

RICE LAKE, WI, USA

WMO: 726467

Lat: 45.419N

Lon: 91.773W

Elev: 337

StdP: 97.34

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54909

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-23.1	-30.0	0.2	-25.3	-27.4	0.3	-22.4	8.9	-5.9	8.1	-7.4	1.0	290	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	31.0	22.6	28.9	21.5	27.5	20.4	24.2	28.4	23.1	27.1	22.0	26.0	4.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.3	26.4	22.1	17.5	25.9	20.9	16.2	24.7	74.8	28.5	70.3	26.9	66.0	25.5	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.8	7.1	-30.6	33.2	3.2	2.0	-32.9	34.7	-34.8	35.8	-36.5	36.9	-38.9	38.3		
(5)			-30.8	26.2	3.2	1.9	-33.0	27.5	-34.9	28.7	-36.7	29.8	-39.0	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-9.8	-8.2	-1.1	6.7	13.0	18.4	20.9	19.3	15.5	8.1	0.9	-6.7		
	DBStd	12.00	7.15	7.07	6.86	5.34	4.61	3.60	3.05	3.08	4.53	4.83	5.50	6.78		
	HDD10.0	2513	613	509	348	124	21	0	0	0	8	96	277	517		
	HDD18.3	4543	872	743	601	349	174	43	11	25	109	318	524	776		
	CDD10.0	1237	0	0	5	26	116	252	338	289	171	37	3	0		
	CDD18.3	225	0	0	0	1	10	44	91	55	22	2	0	0		
	CDH23.3	2041	0	0	1	15	126	386	837	459	202	16	0	0		
	CDH26.7	513	0	0	0	3	31	85	238	105	50	1	0	0		
(14) Wind	WSAvg	2.8	2.7	2.8	3.0	3.5	3.3	2.8	2.3	2.2	2.7	2.9	2.9	2.7		
(15) Precipitation	PrecAvg	837	27	23	49	75	92	118	101	111	89	74	49	29		
	PrecMax	1081	69	54	117	163	162	195	194	230	188	167	169	56		
	PrecMin	601	5	4	18	17	36	43	44	19	20	21	2	7		
	PrecStd	120	16	13	26	36	36	43	39	51	42	41	40	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	8.9	20.1	25.8	30.3	31.3	32.9	32.5	31.1	26.0	18.0	9.0		
		MCWB	3.3	6.1	13.5	15.8	19.3	22.8	24.0	23.2	21.8	17.0	12.4	7.7		
	2%	DB	2.9	6.2	15.2	22.0	27.0	28.9	31.1	29.1	27.8	22.3	14.7	5.7		
		MCWB	1.3	4.0	10.5	12.8	17.3	20.7	23.4	22.1	20.3	16.0	10.7	3.8		
	5%	DB	1.3	3.7	11.6	18.8	24.0	27.4	29.0	27.5	25.9	19.0	12.0	3.6		
		MCWB	0.5	2.0	8.3	10.8	15.8	19.8	22.1	20.7	19.2	14.1	8.3	2.3		
	10%	DB	0.1	1.9	7.8	16.2	21.8	25.9	27.6	26.1	23.3	16.4	8.8	2.1		
		MCWB	-0.6	0.7	5.0	9.3	14.6	18.7	21.0	19.9	17.9	11.9	6.3	1.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	6.5	15.1	17.1	22.0	24.7	26.6	25.0	23.5	20.9	14.2	7.6		
		MCDB	5.5	7.8	18.7	23.3	27.7	29.3	30.2	28.9	27.8	22.1	16.4	8.1		
	2%	WB	1.8	4.1	11.6	14.0	19.1	23.2	24.7	23.5	21.8	17.2	11.5	3.9		
		MCDB	2.9	6.0	13.9	20.1	24.0	26.9	28.9	27.5	26.2	19.9	14.4	4.8		
	5%	WB	0.7	2.2	8.1	11.8	17.4	21.5	23.4	22.3	20.4	14.7	8.3	2.3		
		MCDB	1.4	3.6	11.0	16.8	21.5	25.1	27.5	26.1	24.3	18.4	10.8	3.4		
	10%	WB	-0.8	0.4	5.2	10.1	16.0	20.0	22.3	21.1	18.9	12.5	6.4	0.9		
		MCDB	-0.1	1.5	7.8	15.4	20.0	23.9	26.4	24.5	22.4	15.8	8.8	1.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	10.4	10.1	11.2	11.7	11.0	11.5	11.3	11.7	10.1	8.2	8.0		
		MCDBR	8.8	9.6	13.1	16.2	15.0	12.8	12.9	12.9	13.9	14.4	13.1	9.0		
	5% WB	MCWBR	7.6	7.6	8.2	8.6	7.1	6.1	6.4	6.9	7.7	8.6	8.8	7.5		
		MCWBR	8.3	8.4	12.2	13.9	12.0	10.7	11.4	11.5	11.9	11.6	11.7	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.272	0.310	0.351	0.384	0.385	0.374	0.385	0.359	0.315	0.285	0.261		
		taud	2.385	2.329	2.360	2.364	2.311	2.339	2.389	2.372	2.423	2.532	2.514	2.428		
	Ebn at Noon		873	917	924	908	884	882	887	864	859	860	831	826		
		Edh at Noon	72	93	105	115	125	122	115	113	99	77	64	63		
(44) All-Sky Solar Radiation	RadAvg		1.66	2.77	3.88	4.58	5.32	5.87	6.14	5.14	4.00	2.52	1.54	1.14		
	RadStd		0.18	0.24	0.36	0.46	0.41	0.46	0.24	0.36	0.26	0.30	0.15	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-1.27	N/A	-1.08	-1.31	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page